Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/24/2024
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Project Title:	Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking,& Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	11/09/2017	**End Date:**	03/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2024 per A. Beitman/NASA JSC (Ed., 10/5/23) NOTE: End date changed to 11/08/2023 per A. Beitman/NASA JSC (Ed., 11/3/22) NOTE: End date changed to 11/08/2022 per NSSC information. (Ed. 10/29/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University)		
Grant/Contract No.:	80NSSC18K0276		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking, & Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews," grant NNX15AK73G, due to Principal Investigator move to Northwestern University from Georgia Tech in Fall 2016. We are at the dawn of a new era of human space exploration. Moving beyond low Earth orbit and the relative safety of the International Space Station (ISS) toward near-Earth asteroids and Mars presents previously unimaginable opportunities as well as organizational challenges. One significant challenge is the complexity of the operating environment within which astronauts will work. This complexity will place enormous demands on astronauts, and research is needed that develops concrete countermeasures to mitigate the risks stemming from performance decrements due to inadequate cooperation, coordination, communication, and psychosocial adaptation within a team. Astronauts will push the bounds of human cognitive and social functioning as they multitask across tasks, teams, and tools working toward personal, team, and system goals. This multidisciplinary research project is designed to help them do just that. This investigation into team task switching leverages: (1) agent-based models to understand how task switching behavior and performance-related switching costs occur based on dynamic interplay between independent and interdependent tasks, (2) laboratory experiments conducted in two multiteam systems laboratories (one at Georgia Tech, the other at Northwestern) to test the theoretical model derived from agent-based models, (3) unobtrusive data collection strategies to capture information on the tasks and complex social network structures of ISS crew members, (4) design of interventions to enable adaptive team task switches using virtual experiments, (5) validation of intervention strategies using one of NASA's space analogs, and (6) development of a dashboard decision aid to anticipate and pre-empt dysfunctional task switching. An innovative feature of our investigation is the use of a multidimensional network approach to characterize and model the switches between tasks, tools, teams, and multi-team systems. The combined outputs of this proposed multi-disciplinary project speak directly to NASA Human Research Program's (HRP's) identified team gaps for autonomous, long duration, and/or long distance exploration missions associated with the need to identify (1) key threats to the team (Team Gap 1), (2) countermeasures to support team function (Team Gap 3), (3) validated methods that can be used to maintain team function (Team Gap 5), and (4) psychological and psychosocial factors, measures, and combinations thereof that can be used to compose highly effective crews (Team Gap 8). [Ed. note Sept 2020: HRP Gaps have changed since original proposal--see Human Research Roadmap https://]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous models of task switching revolve around task and individual factors that affect work efficiency as individuals move from task to task throughout their workday. However, individuals not only accomplish tasks individually using one technological tool, they do so as members of multiple teams using a variety of technologies. Our research on team task switching examines factors that affect the decision and ability of individuals to switch between tasks with varying levels of interdependence, as well as discuss the mediators and moderators that determine adaptive switching in today's complicated work environments. This model affords a comprehensive understanding of how individuals adapt to dynamic, environmentally-triggered performance demands requiring them to change tasks, teams, and technologies, and shift back and forth between personal, team, and system goals. Our research will be useful for investigating task switching in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN - the European Council for Nuclear Research), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems.
	Executive Summary Interviews with current and former astronauts as well as reports from astronauts on the International Space Station (ISS) reveal the potential for decrements in crew performance stemming from difficulties in shifting back and forth between independent work and highly interdependent work. For example, ISS crews tend to work for extensive periods of time on independent tasks associated with research projects and other mission-related activities. However, these crewmembers are also expected to switch seamlessly and sometimes spontaneously to interdependent team-based tasks of high criticality and time urgency. We define this as a problem of team task switching. Team task switching impacts the cognitive, motivational, behavioral, and performance effects that result when individuals respond to changing work demands within teams. Changes requiring members to switch tasks, switch teammates, and/or switch tools and technologies deplete attentional resources and make additional cognitive processing demands, which in turn affect the potential for adaptive and seamless task switching. Further, the multiteam structure of NASA requires individuals to regularly shift goal focus in response to dynamic situational requirements. Astronauts often work independently toward a goal, while at other times they work interdependently within a team, and, at yet other times, they work as a part of a large system of teams. Hence, team task switching encompasses both lateral shifts that entail a change in one or more dimensions of work (e.g., task versus tool shifts) as well as vertical shifts that entail a change in the degree of interdependence (e.g., shifting upward from independent to interdependent work, versus shifting downward from interdependent to independent work). This investigation into team task switching allowed us to make contributions to team risk in long distance space exploration missions. Executive Summary of Phase 1 (Agent Based Modeling) & 4 (Virtual Experiments): Through our Phase 1 and Phase 4 efforts, we make four contributions to mitigating team risk in long distance space exploration missions. First, we have developed two new theoretically grounded frameworks. First, a framework of boundary transitions and their effect on individuals working in dynamic teamwork settings. Boundaries in teamwork, described as the psychological limits that define one team as separate from another, have been examined in silos across a broad range of literatures. Our comprehensive model of boundary transitions incorporates a broader perspective of boundary transitions, such as transitions across multiple types of boundaries, as well as dimensions of boundary transitions, such as the strength and meaning of the boundaries traversed, the number of boundaries crossed (transition breadth), and whether the context of the transition is within or between teams (transition context). We leveraged this framework to explore the cognitive implications of boundary transitions (cognitive exertion and expansion) as well as their implications for individual contributions to teamwork. We also developed a theoretical-grounded extension of task

	switching to incorporate the social, team, and situational factors inherent to task switching when work is team based. This extension builds on work in human factors, drawing in organizational theory, and serves as the foundation for the development of a model. Next, using this extended theoretical model, we have developed a computational agent-based model that estimates the stickiness of an ongoing task and the attractiveness of an alternative task to predict the probability that an individual will switch from an ongoing task to an alternative task. This model incorporates the task, social, technological, contextual, and individual factors that affect the perceived stickiness and attractiveness of ongoing versus alternative tasks. Finally, using the computational agent-based model, we have conducted virtual experiments to explain how latency affects task behavior – it reduced the time spent on individual and team tasks, and increased the time spent on multiteam tasks. Executive Summary of Phase 2 (Laboratory Experiments):
Task Progress:	Through our Phase 2 efforts, we contributed to mitigating team risk in long distance space exploration missions by designing and implementing Project RED, a tool for multiteam data collection in analogs. This allowed us to conduct laboratory experiments conducted in two multiteam systems laboratories (one at Georgia Tech, the other at Northwestern) to test the theoretical models derived from agent-based models (from Phase 1) and provide observations and recommendations. Executive Summary of Phase 3 (ISS Observation): Through our Phase 3 efforts, we contributed to mitigating team risk in long distance space exploration missions by conducting unobtrusive data collection to capture information on the tasks and complex social network structures of ISS crew members. We found out that work transitions that involve a gain in autonomy were reported as easier to make by astronauts. Work transitions that involve a loss in autonomy, or conversely, an increase in interdependence, were ranked as being more challenging to astronauts. The easiest transition to make is to go from working with the crew on something to working on a task alone. In contrast, the most challenging transition is to go from a task an astronaut is completing alone to working on a task with mission control. A key message from this data is that the sequencing of work over the course of day matters. Furthermore, overall, astronauts ranked "orienting to the people" as the most difficult, followed by "orienting to the tools," and then by "orienting to the tasks." Interestingly, the rankings change over time. "Orienting to the task" is ranked as "less difficult" relative to the other two as time progresses. "Orienting to the technologies or tools used to do the task" is ranked as "more difficult" relative to the other two as time progresses. "Orienting to the people" has the overall highest ranking, and generally maintains a similar ranking over time relative to tasks and technologies. The data reported by ISS astronauts provides some useful recommendations for designing work in space. Executive Summary of Phase 5 (Analog Study): Through our Phase 4 and 5 efforts, we contributed to the mitigation of team risk in long distance space exploration missions in two ways. First, we have implemented Project RED in short-duration analogs (NASA Human Exploration Research Analog / HERA) to understand which task, social, and situational factors determine time-on-task. We found out that team and situational factors exert strong effects: members whose teams have less similar mental models, those whose teams prioritize their team goal over the multiteam system (MTS) goal, and those working in social isolation and/or under communication delay engage longer on tasks. Time-on-task positively predicts MTS information networks, which in turn positively predict MTS performance when communication occurs with a delay, but not when it occurs in real time. Second, we have implemented Project RED in long-duration analogs (NEK). [Ed. Note: NEK is an acronym for Nazemnyy Eksperimental'nyy Kompleks, which is a multi-compartment facility used as an analog for isolation, confinement, and remote conditions. It is located at the Institute of Biomedical Problems of the Russian Academy of Sciences, Moscow, Russia. The facility is also the location for the Scientific International Research In a Unique terrestrial Station (SIRIUS) missions.] In this study we investigated crew relations over time during the SIRIUS-21 mission. In conjunction with the NEK SIRIUS-21 mission, we assembled two "twin teams" composed of undergraduate students to participate in the same team and multiteam exercises as the SIRIUS-21 crew outside of isolation. Separately, we have investigated how members of MTSs manage relationships within and across multiple types of boundaries, defined by differences in distance (i.e., space vs Earth) and goals (i.e., prevention vs promotion). To understand how such boundaries shape MTS relations, we studied the formation of affective, behavioral, and cognitive ties among the 12-member space exploration MTS who participated in the Project RED multiteam task. Results show MTS members formed a greater proportion of relationships with those who are proximal, than with those with whom they work remotely. Results also show that MTS members were more likely to form external ties with those on teams with the opposite goal focus.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Lungeanu A, DeChurch L, Chan M, Contractor N. "Leading the crew to Mars: Evidence from NASA HERA analog crews." Acta Astronautica. 2025 Aug 16;237:141-50. https://doi.org/10.1016/j.actaastro.2025.08.030 , Aug-2025